ENERGYWAVE, INC. (IN FORMATION) · NEWPORT BEACH, CALIFORNIA

Compartmentalization and Anti-Tamper Posture for the EnergyWave Power Module

Architecture memo — v0.1 (pre-device, architecture level). Draft for counsel review and the due-diligence data room.

Document	Version	Date	Status	Owner
Compartmentalization & AT posture	v0.1	September 12, 2026	DRAFT — for counsel and DD review	Founder, EnergyWave

CLAIMS DISCIPLINE NOTE

This document describes an architecture for protecting information and, conditionally, hardware. It makes no claim that any device exists, works, or will work. Every device-era element below is conditional on a research gate passing under its pre-registered criteria. The mainstream prior on the underlying question remains "no", and nothing here changes that.

1. Purpose, and the goal in one sentence

This memo sets, before there is anything to protect, the architecture by which the program's know-how — and, if any research gate ever produces one, a working power module — is compartmentalized so that a captured unit tells an adversary as little as possible and does nothing for them.

THE GOAL

A unit that falls into the hands of a peer-state laboratory with a fifty-year horizon should be a paperweight: **inert on tamper, and insufficient on its own to rebuild.**

Three reasons to write it now rather than later:

1. **It shapes decisions being made now.** Who sees what, how the lab notebooks are kept, which vendors fabricate which parts, and what gets published under the null-results commitment are all being decided in the first year. Compartmentalization bolted on later means re-doing supplier relationships and re-partitioning documents that were never partitioned.
2. **It is a diligence question.** Deep-tech and defense-adjacent investors ask "how is this protected?" as a matter of course, and any federal path asks it formally through Critical Program Information (CPI) identification and Anti-Tamper (AT) planning.
3. **It has to be written in the company's own register.** The claims discipline is EnergyWave's core asset. A security posture that overpromises — "unbreakable", "un-reverse-engineerable" — would undercut that asset. This memo is honest about what protection can and cannot do, in the same way the research program is honest about the physics.

2. Threat model

WHO

Peer-state laboratories with effectively unlimited time and instrumentation (the working example is a PRC state-affiliated laboratory); well-funded industrial competitors; insiders and departing staff; and supply-chain partners who each see a fragment.

WHAT THEY GET

Case	What the adversary holds	Typical route
A	A finished unit, captured, purchased, or recovered from the field	Loss of a platform; grey-market purchase; theft in transit
B	A unit plus one vendor's fragment of documentation	Compromised fabrication partner; subcontractor breach
C	A whole compartment of know-how, without a unit	Departing employee; insider; contractor
D	Everything published: papers, patents, released data	Public record — by design, under the null-results commitment

WHAT THEY WANT

To characterize (what is it?), to replicate (make one), to operate (make this one work), and to neutralize (know how to defeat it). The layers below are judged by which of these four they deny.

TIME HORIZON AND ITS ONE HARD CONSEQUENCE

The horizon is fifty years. That assumption has a single hard consequence that drives the whole design: any cryptography inside the unit must be assumed breakable within the horizon. Keys buy time; they do not buy fifty years. The "paperweight" property therefore has to rest on physically destroyed state, not on secrets that are merely hard to read.

3. What is honestly possible

Stated plainly, so the architecture is built on what a laboratory can actually do rather than on what a brochure would like to be true.

- **A physical object can always be characterized.** Focused-ion-beam cross-sectioning, electron microscopy, X-ray computed tomography, secondary-ion mass spectrometry and atom-probe tomography will recover the geometry and composition of any lattice, coating or junction. A structure cannot be hidden from a laboratory with fifty years. The design goal is not "they cannot see it"; it is "seeing it is not enough".
- **Structure can be necessary but not sufficient.** Properties that depend on process history — anneal schedules, interface treatments, dopant activation, field or thermal conditioning, defect populations — are measurable in the result but not always recoverable as a recipe. That gap is where a protectable trade secret lives.
- **Anything that must be alive to work can be made to die on tamper.** If a module's function depends on a maintained state — a bias, a field, a temperature, a conditioned configuration — then an enclosure that removes that state on breach turns the unit into a paperweight without needing to hide anything. This is the strongest lever available, and it has to be designed into the first fixture, not the last product.

- **Cryptography imposes cost and delay, not fifty years.** Post-quantum algorithms (NIST FIPS 203, 204 and 205, published 2024) are the right choice for anything with a long service life; the architecture must nonetheless survive their eventual failure.
- **Anti-tamper has a mass budget.** Envelopes, sensors, batteries for tamper circuits and potting all add grams and failure modes. For a power module whose reason to exist is energy density on a mobile platform, every gram of AT counts against the mission. The posture must be sized, not maximal.
- **Secrecy and the publication commitment will collide unless a boundary is drawn.** Publishing null results with data is a program deliverable and stays one. The memo therefore defines a publication boundary (Section 5), not a wall.

4. The layered architecture

Seven layers, numbered from the decision that costs nothing (how knowledge is partitioned) to the one that proves the rest (trying to break it). Layers 0, 3, 4 and 5 apply today. Layers 1, 2 and 6 are device-era and conditional.

Layer 0 — Partition the knowledge

Sort every element of the program into three bins, and keep them apart.

Bin	What it holds	Assumption	Protection
(a) Imageable	Geometry, materials, layer stack, packaging	Recoverable from a captured unit	Patent where counsel advises, or accept as public
(b) Process	Recipes, conditioning schedules, fixture designs, calibration, yield learnings	Not recoverable from the object alone	Trade secret — the crown-jewel bin; never in one document
(c) State	Whatever must be present for function and is destroyed on breach — in the program's own vocabulary, the conditioned core	Erasable by design	Physical AT (Layer 1); the paperweight mechanism

The rule: no single person, vendor, or document holds bin (b) and bin (c) together with the assembled bin (a). An adversary in Case A has (a). An adversary in Case B has (a) plus a slice of (b). Neither has (c), and neither has enough of (b) to run the process.

Layer 1 — Physical anti-tamper (device era, conditional)

A tamper-evident and tamper-responsive enclosure in the sense used by FIPS 140-3 at Security Levels 3 and 4: a detection envelope around the critical volume, environmental-failure protection that treats voltage and temperature excursions as attacks, and zeroization of secrets on breach. Two grades of response, deliberately separated:

- **Delay:** potting, active mesh, and sensor envelopes. These cost a skilled laboratory days to weeks and are worth having, but they are not what defeats a fifty-year attacker.
- **Kill:** a keep-alive condition for the core whose interruption irreversibly de-conditions it. This is the paperweight mechanism proper. It works only if the core's function genuinely depends on a

maintained state — which is a design requirement to carry into every fixture from Gate 1 onward, not a feature to add at productization.

The reference frameworks are the Department of Defense Anti-Tamper directive (DoDD 5200.47E) and the CPI identification and protection instruction (DoDI 5200.39). Both would apply formally on any federally funded path; adopting their vocabulary now costs nothing and shortens that path later.

Layer 2 — Cryptographic platform binding (device era, conditional)

This extends the platform-bound control architecture already described in the DoD white paper: per-unit keys held in a secure element with zeroization on tamper; authenticated firmware; drive and control parameters encrypted at rest and released only to an attested controller; no plaintext parameter tables anywhere in the unit. Primitives should be post-quantum (FIPS 203/204/205) given the service life. The role of this layer is operational control and cost imposition — a stolen unit will not run for its new owner even before anyone opens it — not the final barrier. Section 2 explains why it cannot be.

Layer 3 — Split manufacturing and supplier compartmentalization

No fabrication vendor sees the full stack: patterning at one vendor, coatings at another, assembly and conditioning in-house. This is the split-manufacturing model the U.S. trusted-foundry community developed (IARPA's Trusted Integrated Chips program is the reference). Vendor drawings carry code names per sub-assembly; purchase orders and drawings for one sub-assembly never reference another; the crown-jewel step never leaves the company. Applies from the first fixture order.

Layer 4 — Information compartmentalization

- **Three-envelope rule:** no single document contains the full recipe. Lab notebooks are segmented by bin.
- **Measurement data is kept apart from process data.** The two-question-protocol results (net source? colder sink?) are what gets published; the process that produced the sample is not in the same file.
- **Role-based access,** with advisors seeing the compartment they advise on. NDAs are matched to compartments; departure procedures return and revoke by compartment.
- **This is also a legal element, not a nicety.** Under the Defend Trade Secrets Act the definition of a trade secret includes that the owner has taken reasonable measures to keep it secret. A compartmentalization program is the evidence of those measures.

Layer 5 — Legal and regulatory posture (questions for counsel)

None of the following is legal advice; it is the question list for K&L Gates, organized by bin.

- **Trade secret versus patent, per bin.** A patent discloses at eighteen months and expires at twenty years; a trade secret lasts indefinitely but gives no protection against independent discovery or reverse engineering — which is precisely why Layers 1 to 3 exist. Bin (a) leans patent; bin (b) leans trade secret; bin (c) is protected physically.
- **Invention Secrecy Act exposure.** A patent application on energy technology with defense relevance can receive a secrecy order (35 U.S.C. §§ 181–188). Filing strategy should anticipate that.
- **Export controls.** Classification of the program's technical data under the EAR (Commerce/BIS) or, if it were ever a defense article, the ITAR (State/DDTC). The deemed-export rule means releasing

controlled technology to a foreign national inside the United States is itself an export — directly relevant to an advisory network with international members and to any university partnership.

- **CFIUS.** Under FIRRMA, foreign investment in a U.S. business involved in critical technologies can trigger mandatory filings. This bears on how the raise is structured and who is admitted to it.
- **Federal information rules.** NIST SP 800-171 for controlled unclassified information and the CMMC program apply the moment any federal work is contracted; the SAM.gov registration already on the federal to-do list is the on-ramp.

Layer 6 — Operational proof: the paperweight test (device era, conditional)

Protection is validated the same way the physics is: by trying to break it. Make "can a friendly laboratory make it run, or rebuild it from a unit?" a deliverable — a red-team characterization exercise producing a time-to-characterize and time-to-replicate report, run by a party with no other involvement in the program. Slot it into the gate structure as a Gate 3 deliverable, with pass and kill criteria pre-registered like everything else.

5. What each capture case yields, and what the layers deny

Case	Adversary can…	Denied by	Residual risk
A — unit only	Characterize bin (a) fully	Layer 1 kill (state gone); Layer 2 (will not run); Layer 0 (no process)	Geometry and materials become known — assumed from the start
B — unit + vendor fragment	Characterize (a); learn one sub-assembly's process	Layer 3 (fragment is one of several); Layer 0	Narrows the search for the rest of bin (b)
C — one compartment, no unit	Reproduce that compartment	Layer 4 (no compartment is sufficient); Layer 5 (DTSA remedies)	Insider with cross-compartment access — the governance failure mode
D — public record	Read the physics and the null results	Publication boundary (Section 6)	None by design; this is the program's commitment

Read across the rows: no single case lets an adversary operate a captured unit or rebuild one, provided the kill mechanism in Layer 1 is real. That is the load-bearing assumption of the whole posture, which is why Layer 6 exists to test it and why the design requirement has to enter at the first fixture.

6. What changes now, before any device exists

1. **Partition from day one.** Lab notebooks and fixture designs are kept by bin (a)/(b)/(c) starting with Gate 1 preparation.
2. **Write the publication boundary.** Publish measurements, protocols, null results and the physics; protect process, conditioning and fixture detail. One page, so the null-results commitment and the trade-secret program cannot collide by accident.

3. **Adopt the vendor policy.** Split manufacturing for any fabrication order; code-named sub-assemblies; the crown-jewel step in-house.
4. **Set access by role.** Advisors and hires see the compartment they work in; NDAs and departure procedures follow the compartments.
5. **Carry the kill requirement into fixture design.** Every fixture from Gate 1 is designed so that the conditioned state, if one exists, depends on a maintained condition that can be interrupted.
6. **Take the Layer 5 question list to counsel.** Trade-secret program, filing strategy and secrecy-order exposure, export classification and deemed-export handling, CFIUS considerations for the raise.

7. Implications for the raise materials

Deliverable	What goes in	What stays out
Investor deck	One slide, "IP and security posture": partition; paperweight on tamper; split manufacturing; counsel-led legal posture	Mechanisms, vendors, anything that reads as a claim the device exists
Technical prospectus	One conditional paragraph under the research program	Layer detail
DD data room	This memo, v0.1, marked architecture-level and pre-device; counsel's response when received	Nothing — this is the right place for the full document
Public website	At most one posture line	Specifics — they signal targets and imply a device

8. Open questions for counsel

1. Design of a trade-secret program that documents "reasonable measures" from the first month of operations.
2. Patent versus trade secret allocation by bin, and Invention Secrecy Act exposure for any energy-related filing with defense relevance.
3. Export classification of the research program's technical data, and deemed-export handling for international advisors and any university partner.
4. CFIUS/FIRRMA considerations for the raise: which investors, which structures, whether a declaration could be mandatory.
5. CPI and Anti-Tamper obligations that would attach to any federal funding, and what to prepare in advance.
6. NDA templates and departure procedures aligned to compartments rather than to the company as a whole.

9. Version and next steps

Version	Date	Change
v0.1	September 12, 2026	First architecture-level draft. Pre-device. Written for counsel review and the data room.
v0.2 (planned)	—	Fold in counsel's answers to Section 8; attach the one-page publication boundary; add the paperweight test to the Gate 3 criteria.

REFERENCES (FRAMEWORKS NAMED IN THIS MEMO)

DoDD 5200.47E, Anti-Tamper (AT), Department of Defense, 2015 as amended. · DoDI 5200.39, Critical Program Information (CPI) Identification and Protection Within RDT&E, 2015 as amended. · FIPS 140-3, Security Requirements for Cryptographic Modules (physical security Levels 1–4); FIPS 140-2 certificates retire in September 2026. · NIST FIPS 203 (ML-KEM), 204 (ML-DSA), 205 (SLH-DSA), August 2024. · Defend Trade Secrets Act of 2016, 18 U.S.C. § 1836 et seq.; definition at § 1839(3). · Invention Secrecy Act, 35 U.S.C. §§ 181–188. · Export Administration Regulations, 15 C.F.R. Parts 730–774 (deemed exports at § 734.13); International Traffic in Arms Regulations, 22 C.F.R. Parts 120–130. · FIRRMA (2018) and CFIUS regulations, 31 C.F.R. Part 800. · NIST SP 800-171; CMMC, 32 C.F.R. Part 170. · IARPA Trusted Integrated Chips (TIC) program, split manufacturing.